Microeconomic Theory: TA Session

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April 24, 2026

Reminder

There is still a class on Monday, April 27

The final exam will be on May 11th, 9am-11am

Risk aversion

Reminder. If a person's utility function is $u(w)$:

- He is risk averse if $u''(w) < 0$.
- He is risk neutral if $u''(w) = 0$.
- He is risk loving if $u''(w) > 0$.

Insurance

There are two states: Good (G) and Bad (B)

- Good state happens with probability (1-p), bad state happens with probability p
- Initial wealth: $w_G = w_0,\ w_B = w_0 - x$

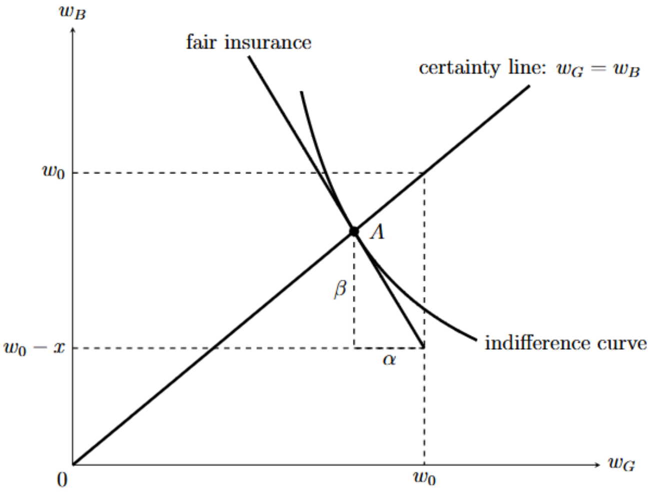
An insurance company offers insurance

- Insurance premium: α (good-state wealth is now $w_G = w_0 - \alpha$)
- Net indemnity: β (bad-state wealth is now $w_B = w_0 - x + \beta$)

The insurance policy satisfies $-(1-p)\alpha + \lambda p \beta = 0$

- $\lambda = 1$: fair insurance
- $\lambda > 1$: unfair insurance

Insurance



w_B
fair insurance
certainty line: $w_G = w_B$
w_0
A
β
$w_0 - x$
indifference curve
α
0
w_0
w_G

Fair insurance

The consumer's problem is to choose (α, β) to maximize:

$$(1-p)U(w_0-\alpha)+pU(w_0-x+\beta)$$
$$\text{s.t. } -(1-p)\alpha+\lambda p\beta=0$$

Proposition 1. Under fair insurance ($\lambda = 1$), a risk-averse consumer will take out full insurance (so that $w_G = w_B$).

Proof: The insurance policy satisfies $-(1-p)\alpha + p\beta = 0$, so the slope of the fair insurance line is $\frac{d\beta}{d\alpha} = -\frac{1-p}{p}$.

Consumer's expected utility is $(1-p)U(w_0-\alpha)+pU(w_0-x+\beta)$. Then, the slope of indifference curve is $-MRS_{G,B} = -\frac{MU_G}{MU_B} = -\frac{(1-p)U'(w_0-\alpha)}{pU'(w_0-x+\beta)}$

At the optimal tangential point, the two slopes are equal:

$-\frac{1-p}{p} = \frac{(1-p)U'(w_0-\alpha)}{pU'(w_0-x+\beta)} \;\Rightarrow\; w_0-\alpha = w_0-x+\beta,\; w_G = w_B.$

Alternative proof of proposition 1

Alternative algebraic proof without the help of a graph:

Rewrite the fair insurance condition as $\alpha = \frac{p}{1-p}\beta$. Expected utility:
$(1-p)U(w_0-\alpha)+pU(w_0-x+\beta) = (1-p)U(w_0-\frac{p}{1-p}\beta)+pU(w_0-x+\beta)$.
Taking derivative with regard to β:
$(1-p)U'(w_0 - \frac{p}{1-p}\beta)(-\frac{p}{1-p}) + pU'(w_0 - x + \beta) = 0$
$\Rightarrow \; U'(w_G) = U'(w_0 - \frac{p}{1-p}\beta) = U'(w_0 - x + \beta) = U'(w_B)$
$\Rightarrow w_G = w_B$, the consumer takes out full insurance.

Unfair insurance

$$(1-p)U(w_0-\alpha)+pU(w_0-x+\beta)$$
$$\text{s.t. } -(1-p)\alpha+\lambda p\beta=0$$

Proposition 2. Under unfair insurance, a risk-averse consumer will take out partial insurance ("with a minimum deductible", in Karni's words)

Proof: Rewrite the insurance constraint as $\alpha=\frac{p}{1-p}\lambda\beta$, then plug this into the expected utility function:

$$(1-p)U(w_0-\frac{p}{1-p}\lambda\beta)+pU(w_0-x+\beta)$$

Taking the derivative with regard to β:

$$(1-p)U'(w_0-\frac{p}{1-p}\lambda\beta)(-\frac{p}{1-p}\lambda)+pU'(w_0-x+\beta)=0$$
$$\lambda U'(w_0-\frac{p}{1-p}\lambda\beta)=U'(w_0-x+\beta)$$
$$U'(w_G)<U'(w_B)\ \Rightarrow w_G>w_B, \text{ because risk aversion means } U''(w)<0$$

Unfair insurance

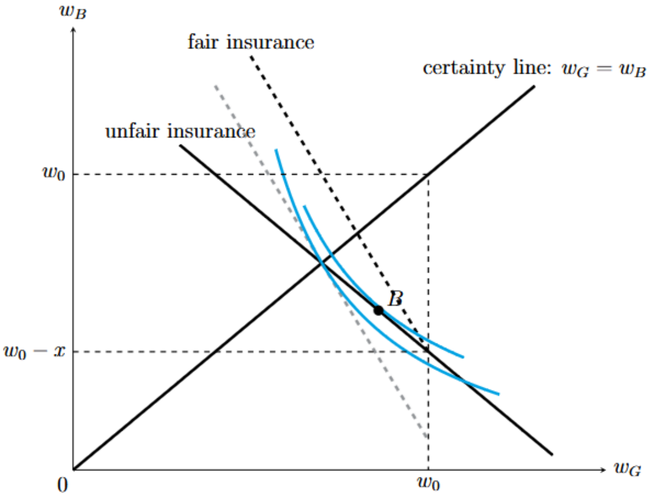


Figure: Partial insurance if insurance is unfair

Insurance and the degree of risk aversion

Proposition 3. Other things being equal, a more risk-averse person will take out a larger amount of insurance ("will have a smaller deductible", in Karni's words)

Proof is omitted. It is intuitive that the more risk-averse a person is, the more insurance he would like to have.

Insurance - exercise

Consider a consumer facing uncertainty. In the good state, which happens with probability 90% his wealth is $w_G = 10,000$. In the bad state, which happens with probability 10%, his wealth is $w_B = 1000$. Suppose he can buy insurance with a insurance premium of α and a net indemnity of β.

1. What is the relationship between α and β under fair insurance?

   Answer: Bad state happens with $p = 0.1$. Under fair insurance, $-(1-p)\alpha + p\beta = 0 \;\Rightarrow\; -0.9\alpha + 0.1\beta = 0 \;\Rightarrow\; \beta = 9\alpha$.

2. Suppose the consumer's utility function is $u(w) = \ln w$. Under fair insurance, what is the optimal amount of insurance he buys?

   Answer:

   His expected utility is $0.9 \times \ln(10000 - \alpha) + 0.1 \times \ln(1000 + \beta)$
   $= 0.9 \times \ln(10000 - \alpha) + 0.1 \times \ln(1000 + 9\alpha)$.

   First-order condition with regard to α: $-\frac{0.9}{10000-\alpha} + \frac{9\times 0.1}{1000+9\alpha} = 0$

   $\Rightarrow\; \alpha = 900,\; \beta = 8100$. He will be fully insured.

Insurance - exercise

Consider a consumer facing uncertainty. In the good state, which happens with probability 90% his wealth is $w_G = 10,000$. In the bad state, which happens with probability 10%, his wealth is $w_B = 1000$. Suppose he can buy insurance with a insurance premium of α and a net indemnity of β.

3. Now suppose that, instead of fair insurance, the insurance company sets $\beta = 4\alpha$. What is the optimal amount of insurance he buys?

   Answer:

   His expected utility is $0.9 \times \ln(10000 - \alpha) + 0.1 \times \ln(1000 + \beta)$
   $= 0.9 \times \ln(10000 - \alpha) + 0.1 \times \ln(1000 + 4\alpha)$.

   First-order condition: $-\frac{0.9}{10000-\alpha} + \frac{4\times 0.1}{1000+4\alpha} = 0$

   $\Rightarrow \; \alpha = 775, \; \beta = 3100.$

   This is partial insurance, because $w_G = 10000 - \alpha = 9225$,
   $w_B = 1000 + 4\alpha = 4100$.

Insurance under asymmetric information

Under asymmetric information, the setup is as follows:

There are two types of consumers

- High-risk type: probability of bad state is p^H
- Low-risk type: probability of bad state is p^L
- $p^H > p^L$

Each individual knows his own type; insurer only knows that the proportion of high-risk people in the population is $\lambda \in (0, 1)$.

Population's average probability of bad state: $\bar{p} = \lambda p^H + (1 - \lambda) p^L$

Definition of equilibrium

In this problem, an equilibrium (called Rothschild-Stiglitz equilibrium) is a set C of insurance policies offered such that:

1. Every policy $(\alpha, \beta) \in C$ is (weakly) profitable
2. For every other policy $(\alpha', \beta') \notin C$, if it is to be offered along with existing policies in C, then (α', β') is not profitable.

In words: in equilibrium, there is no incentive to withdraw an existing insurance policy or offer a new policy

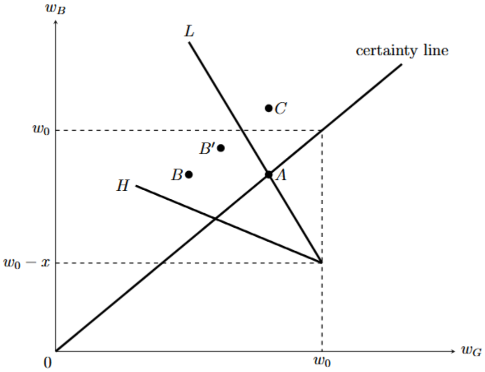
Benchmark case: symmetric information

Before going into asymmetric information, we first consider the "benchmark" case of symmetric information where the insurer knows each individual's type, and then compare asymmetric info solution with this benchmark

Consider the low-risk type.

Any policy below the L line (such as point B) cannot be an equilibrium, because a new policy slightly to the northeast of B can be offered (B') that is better for consumers and also profitable for insurer

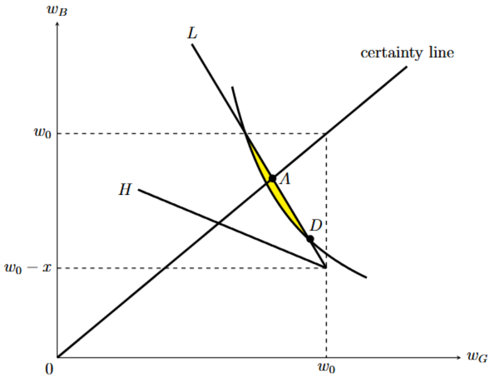
Any policy above the L line (such as point C) cannot be an equilibrium, because the insurer makes a loss



Benchmark case: symmetric information

Any policy on the L line that is not full insurance (such as point D) cannot be an equilibrium, because we can offer a new policy between D's indifference curve and the L line (the yellow region in the graph below) that is better for the consumer and is also profitable for the insurer

So, the only possible equilibrium is full insurance (point A)



Benchmark case: symmetric information

The same argument applies for the high-risk type.

Then, under symmetric information, the equilibrium is to offer both types full insurance.

This is welfare-maximizing (see page 5-6 of these slides)

Asymmetric information: no pooling equilibrium

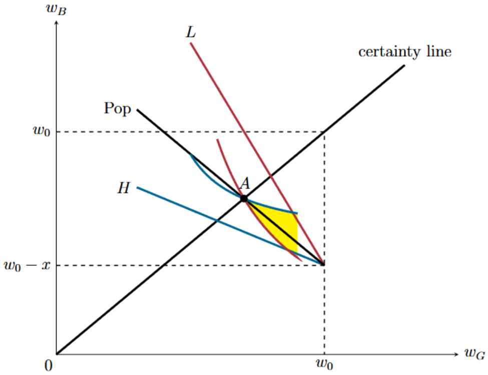
Under asymmetric information,

Proposition. There is no "pooling equilibrium" where both types are offered the same policy

Any policy below or above the "Population" line cannot be the equilibrium.

Any policy on the population line cannot be the equilibrium either. Consider such a policy *A*. We draw both type's indifference curves that pass through *A*.

Then we can offer a new policy in the "southeastern" region between the two ICs (the yellow region) that attracts the low-risk type and is profitable.

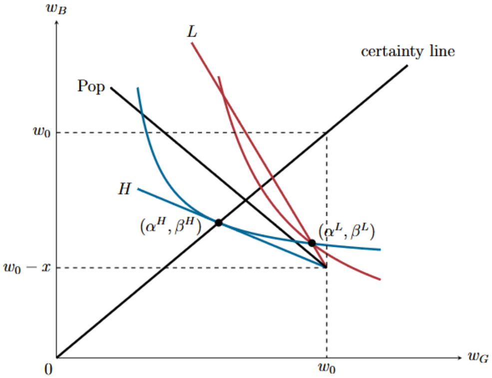


Asymmetric information: separating equilibrium

If a separating equilibrium exists, the insurer will offer full insurance (α^H, β^H) to the high-risk type. Reasoning is the same as before.

Draw the high-risk type's IC that passes through (α^H, β^H). This will intersect the L line at (α^L, β^L) — this is the low-risk type's equilibrium insurance policy.

You can use the definition of equilibrium to check why this is an equilibrium, and why other policies cannot be an equilibrium.



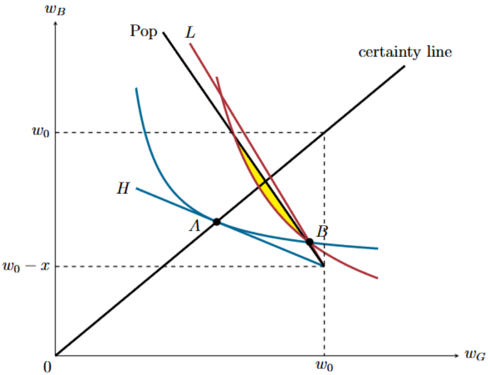
Welfare: for the low-risk type, this gives less utility than full insurance

Asymmetric information: separating equilibrium

However, if λ is very close to 0, then there does not exist a separating equilibrium.

To see this, notice that with the insurance policies of the previous page, we can offer a new insurance policy between the low-risk type's IC and the population line (the yellow region). This new policy will attract everyone, and earn a positive profit.

Intuition: with a small λ, the insurer can afford to subsidize the loss from the few high-risk people using the profits made from the many low-risk people.



Asymmetric information - exercise

Suppose that the low-risk type ends up in the bad state at the probability of $p^L = 10\%$, and the high-risk type has $p^H = 20\%$. Both types has $w_G = 10,000$ in good state and $w_B = 1000$ in bad state. Their utility function is $u(w) = \ln w$. In the population, $\lambda = 50\%$ of all people are high-risk. A perfectly competitive insurance market is offering insurance to these two types.

1. Find the equilibrium under symmetric information.

   Answer: Under symmetric information, the equilibrium is full insurance for everyone.

   For the high-risk type: $-(1-p^H)\alpha^H + p^H\beta^H = 0 \;\Rightarrow\; \beta^H = 4\alpha^H$.

   Full insurance: $w_G = w_B \;\Rightarrow\; 10000 - \alpha^H = 1000 + \beta^H$,

   Solving, we get: $\alpha^H = 1800,\ \beta^H = 7200$.

   For the low-risk type: $-(1-p^L)\alpha^L + p^L\beta^L = 0 \;\Rightarrow\; \beta^L = 9\alpha^L$.

   Full insurance: $w_G = w_B \;\Rightarrow\; 10000 - \alpha^L = 1000 + \beta^L$,

   Solving, we get: $\alpha^L = 900,\ \beta^L = 8100$.

Asymmetric information - exercise

2. Under asymmetric information, assume that the separating equilibrium exists. Find the separating equilibrium.

   Answer: For the high-risk type, the (separating) equilibrium is still full insurance: $\alpha^H = 1800, \beta^H = 7200$.

   High-risk type's expected utility: $EU^H = (1 - p^H)u(w_G) + p^H u(w_B) = 0.8 \times \ln(8200) + 0.2 \times \ln(8200) = \ln(8200) \approx 9.012$.

   Low-risk type's equilibrium insurance policy lies on the intersection of high-risk type's indifference curve and low-risk type's fair insurance line.

   High-risk type's IC has expected utility at ln(8200). Low-risk type's fair insurance line is $-0.9\alpha^L + 0.1\beta^L = 0 \;\Rightarrow\; \beta^L = 9\alpha^L$.

   Recall also that $w_G^L = 10000 - \alpha^L, \; w_B^L = 1000 + \beta^L$.

   At the intersection, the expected utility for a high-risk type would have been ln(8200):

   $EU^H = 0.8\ln(10000 - \alpha^L) + 0.2\ln(1000 + \beta^L) =$
   $0.8\ln(10000 - \alpha^L) + 0.2\ln(1000 + 9\alpha^L) = \ln(8200)$.

   Solving with a calculator, we get $\alpha^L = 367.35, \; \beta^L = 3306.15$.